Research Proposal: A Top-Down Temporal Hierarchy for LeWM

With Hybrid CEM–ID Planning

Abstract

We propose an extension to the LeWorldModel (LeWM) that addresses its primary limitation: the compounding of prediction errors over long planning horizons. Current world models either rely on a *bottom-up* hierarchy—progressive abstraction from pixels through chunks to episodes—or flat autoregressive rollout, both of which accumulate errors step-by-step. Our *Top-Down Temporal Hierarchy* instead first extrapolates a distant future anchor state, then interpolates backward to fill intermediate waypoints, before resolving fine-grained reactive steps. This "predict the end, fill the middle" paradigm ensures every prediction is constrained by a higher-level goal, dramatically reducing error propagation. Crucially, we propose a *hybrid inference scheme*: Levels 2 and 3 are solved analytically via Inverse Dynamics (ID) networks, while Level 1 retains a short-horizon Cross-Entropy Method (CEM) optimizer anchored by the mid-range waypoint above it. This yields a planner that is simultaneously more accurate over long horizons and significantly cheaper to optimize than flat CEM.

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1 Introduction: Reimagining State Abstraction

Training a world model end-to-end from raw pixels introduces the challenge of *representation collapse*, wherein the encoder maps every input to a constant vector, trivially satisfying the prediction objective. LeWM addresses this with a Joint Embedding Predictive Architecture (JEPA) regularized by the Sketched Isotropic Gaussian Regularizer (SIGReg), achieving state-of-the-art visual planning at a fraction of competing models' computational cost.

Despite this progress, flat autoregressive rollout—the planning backbone of base LeWM—accumulates prediction errors monotonically with horizon length H. Each step prediction is made "blind" to the long-range goal; the only feedback is the terminal cost, which CEM must bridge across the entire sequence. For manipulation tasks or navigation requiring hundreds of steps, this becomes prohibitively unreliable.

Existing hierarchical world models adopt a **bottom-up** strategy: separate encoder towers construct increasingly abstract latent representations (pixels → chunks → episodes), each with its own prediction head. This introduces two compounding failure modes: (1) independent representation collapse can occur at each level, requiring separate regularization; and (2) cross-level prediction errors stack at the abstraction boundaries, exacerbating the original problem.

We propose instead a **Top-Down Temporal Hierarchy** that retains the flat, single-encoder latent space of base LeWM. The hierarchy is expressed *entirely through temporal distance*: a strategic predictor jumps to the distant future, a tactical predictor interpolates a

midpoint, and a reactive predictor resolves individual steps. This design eliminates abstraction-boundary errors, requires SIGReg only once, and enables a natural hybrid inference scheme where the computationally expensive search (CEM) is confined to the shortest, most tractable horizon.

2 Background: Base LeWM

2.1 Architecture

Base LeWM consists of two learned components operating in a shared latent space:

- **Encoder** enc_θ (ViT-Tiny, $\approx$ 5M parameters): maps a raw pixel frame o_t to a compact latent vector $z_t \in \mathbb{R}^d$.
- **Predictor** pred_ϕ (ViT-Small transformer, $\approx$ 10M parameters): given the current latent z_t and action a_t, predicts the next latent state $\hat{z}_{t+1}$.

2.2 Training Objective

Training minimizes a two-term loss with a single hyperparameter λ:

$$\mathcal{L} = \underbrace{\|\hat{z}_{t+1} - z_{t+1}\|_2^2}_{\text{prediction loss}} + \lambda \cdot \underbrace{\text{SIGReg}(Z)}_{\text{anti-collapse}} \tag{1}$$

SIGReg prevents collapse by projecting the batch of embeddings Z onto many random directions and applying the Epps–Pulley normality test to each 1D projection. By the Cramér–Wold theorem, matching all marginal 1D distributions to a Gaussian is equivalent to matching the full joint distribution to an isotropic Gaussian.

2.3 Planning: Flat CEM

At test time, LeWM must find an action sequence $a_{1:H}$ that brings the agent from start observation o_1 to goal observation o_g. This is formulated as:

$$a^*_{1:H} = \arg\min_{a_{1:H}} \|\hat{z}_H - z_g\|_2^2 \tag{2}$$

where $\hat{z}_H$ is obtained by autoregressively rolling out the predictor in latent space. Since this cost is not differentiable with respect to $a_{1:H}$ in closed form, LeWM uses the Cross-Entropy Method (CEM): a sampling-based optimizer that iteratively concentrates the action distribution around low-cost candidates.

Limitation. As H grows, the autoregressive rollout accumulates errors multiplicatively, and the CEM search space grows linearly, rapidly degrading planning reliability.

3 Proposed Architecture: Top-Down Temporal Hierarchy

3.1 Flat State Encoding

The foundation of the hierarchy is unchanged from base LeWM: a single shared encoder maps every frame into the same latent space, regardless of when in the trajectory it occurs:

$$z_t = \text{enc}_\theta(o_t), \quad \forall\, t \tag{3}$$

This is a critical design decision. Unlike bottom-up hierarchies, there are no abstractor networks and no separate embedding spaces. All states—present, midpoint, and distant future—occupy identical positions in the same $\mathbb{R}^d$ geometry. This enables the ID models to operate purely over temporal distance rather than over qualitatively different representations.

3.2 Level 3 — Strategic Extrapolation

Prediction begins at the highest level. The Level 3 predictor takes the current state z_t and a high-level macro-action $\tilde{a}_t^{(3)}$ to predict a state k_2 steps into the future, establishing a *long-range anchor*:

$$\hat{z}_{t+k_2}^{(3)} = \text{pred}^{(3)}\Big(z_t,\ \tilde{a}_t^{(3)}\Big) \tag{4}$$

The macro-action $\tilde{a}_t^{(3)}$ is produced by the **Strategic Inverse Dynamics** model, which infers long-range intent by analyzing the state pair (z_t, z_{t+k_2}):

$$\tilde{a}_t^{(3)} = \text{ID}^{(3)}(z_t,\ z_{t+k_2}) \tag{5}$$

Semantic interpretation. Level 3 encodes coarse, task-level intent—"move the object from center to right," "cross the room," "reach the goal configuration." It has no concept of joint angles or intermediate positions; it only models what the world looks like before and after a meaningful high-level event.

3.3 Level 2 — Tactical Interpolation

With two anchor points now known in latent space—the current state z_t and the distant future $\hat{z}_{t+k_2}^{(3)}$—the Level 2 predictor *interpolates* a midpoint state at step k_1 ($t < t{+}k_1 < t{+}k_2$):

$$\hat{z}_{t+k_1}^{(2)} = \text{pred}^{(2)}\Big(z_t,\ \hat{z}_{t+k_2}^{(3)},\ \tilde{a}_t^{(2)}\Big) \tag{6}$$

The mid-range macro-action is produced by the **Tactical Inverse Dynamics** model:

$$\tilde{a}_t^{(2)} = \text{ID}^{(2)}(z_t,\ z_{t+k_1}) \tag{7}$$

The Level 3 anchor acts as a gravitational constraint on this prediction: the midpoint cannot drift arbitrarily far off course because the model already knows where the trajectory must pass through at $t{+}k_2$.

Semantic interpretation. Level 2 encodes sub-goal reasoning—"hover the hand above the cup before grasping," "reach the doorway between rooms," "get within attack range." It understands waypoints and staging, but not individual motor commands.

3.4 Level 1 — Reactive Step Prediction

Finally, with both the current state z_t and the near-future midpoint $\hat{z}_{t+k_1}^{(2)}$ known, Level 1 resolves the immediate next frame using the ground-truth physical action a_t:

$$\hat{z}_{t+1}^{(1)} = \text{pred}^{(1)}\Big(z_t,\ \hat{z}_{t+k_1}^{(2)},\ a_t\Big) \tag{8}$$

Level 1 intentionally bypasses an ID model. Physical actions are low-level and noisy—joint torques, pixel-level displacements—making them difficult to infer reliably from state pairs alone. Instead, at inference time, Level 1 actions are the *only* remaining optimization variable, handled by a short-horizon CEM (see Section 6).

Semantic interpretation. Level 1 encodes pure motor control—"extend joint 3 by 2 degrees," "move 0.3 cm along the x-axis," "press button A." It has no concept of the final goal; it only knows where it is and where it must be within the next k_1 steps.

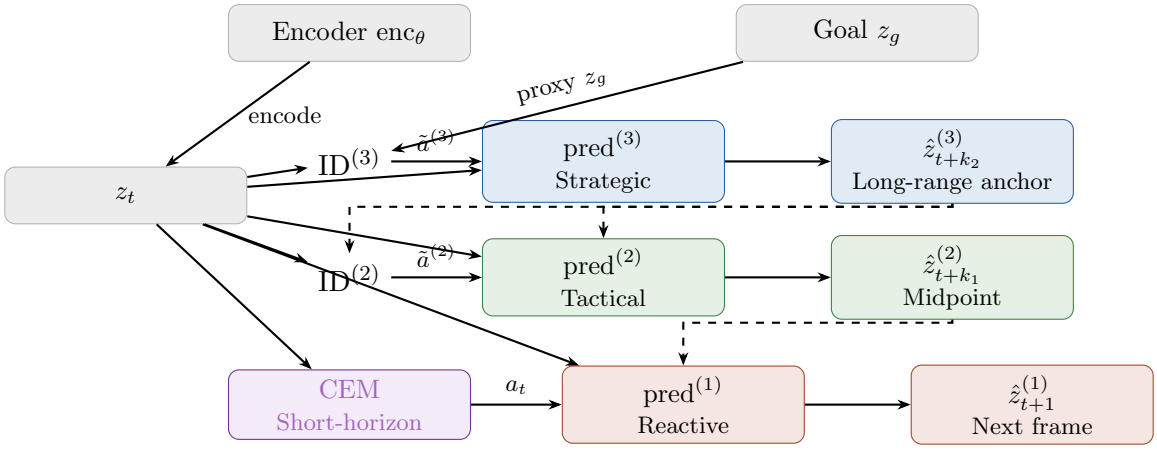

Figure 1: Top-Down Temporal Hierarchy. Dashed arrows denote top-down guidance from higher-level anchors. CEM operates only at Level 1, over a short horizon anchored by $\hat{z}^{(2)}_{t+k_1}$.

3.5 Full Architecture Diagram

4 Temporal Inverse Dynamics (ID) Models

4.1 Design Rationale

Because all states reside in a single unified latent space, the ID models can extract intent purely by analyzing the *geometric displacement* between two states, without needing to parse qualitatively different abstraction levels. The only varying quantity across levels is the temporal distance separating the state pair.

4.2 Strategic ID (Level 3)

Infers long-range intent from the current state and the target distant-future state:

$$\tilde{a}^{(3)}_t = \text{ID}^{(3)}(z_t,\ z_{t+k_2}) \tag{9}$$

At planning time, the true z_{t+k_2} is unavailable; the goal embedding z_g serves as its proxy, directing the extrapolator toward the desired terminal configuration.

4.3 Tactical ID (Level 2)

Infers mid-range intent from the current state and the target midpoint:

$$\tilde{a}^{(2)}_t = \text{ID}^{(2)}(z_t,\ z_{t+k_1}) \tag{10}$$

4.4 Reactive Level: No ID Model

Level 1 bypasses an ID model by design. Low-level physical actions (joint torques, pixel-level displacements) are inherently ambiguous when inferred from state pairs alone—many different action sequences can produce visually similar state transitions. CEM is more reliable here because it evaluates *actual rollout cost* rather than approximating actions from endpoints.

5 Training Objective

5.1 Forward Prediction Losses

Each level is supervised against the true future state encoded by the base encoder:

$$\mathcal{L}_{\text{pred}}^{(3)} = \left\| \hat{z}_{t+k_2}^{(3)} - z_{t+k_2} \right\|_2^2 \tag{11}$$

$$\mathcal{L}_{\text{pred}}^{(2)} = \left\| \hat{z}_{t+k_1}^{(2)} - z_{t+k_1} \right\|_2^2 \tag{12}$$

$$\mathcal{L}_{\text{pred}}^{(1)} = \left\| \hat{z}_{t+1}^{(1)} - z_{t+1} \right\|_2^2 \tag{13}$$

5.2 Anti-Collapse Regularization

Because all levels share the same encoder, SIGReg is applied *exactly once* to the base encoder's output batch Z:

$$\mathcal{L}_{\text{reg}} = \lambda \, \text{SIGReg}(Z) \tag{14}$$

This is a significant simplification over bottom-up hierarchies, which require separate anti-collapse mechanisms at each abstraction level.

5.3 Latent Action Regularization

To prevent the ID models from degenerate shortcuts (e.g., copying the target state directly into the action representation), an L_2 penalty is applied to the macro-action vectors:

$$\mathcal{L}_{\text{act}} = \alpha \left(\left\| \tilde{a}_t^{(2)} \right\|_2^2 + \left\| \tilde{a}_t^{(3)} \right\|_2^2 \right) \tag{15}$$

5.4 Total Objective

The model is optimized end-to-end via a single consolidated loss:

$$\boxed{\mathcal{L}_{\text{total}} = \sum_{l \in \{1, k_1, k_2\}} \| \hat{z}_{t+l} - z_{t+l} \|_2^2 + \lambda \, \text{SIGReg}(Z) + \alpha \sum_{l \in \{2,3\}} \left\| \tilde{a}^{(l)} \right\|_2^2} \tag{16}$$

This is only marginally more complex than the base LeWM objective (two terms, one hyperparameter), now extended with two additional prediction heads and a two-hyperparameter regularizer.

6 Inference and Planning

This section is the primary extension over the original proposal. We introduce a **Hybrid CEM–ID Planning** scheme that exploits the complementary strengths of the ID networks and CEM: the ID networks handle high-level action inference deterministically, while CEM operates exclusively over the low-level physical action space, guided by the anchors produced above it.

6.1 Overview

6.2 Phase 0: Encode Start and Goal

Both the current observation and the goal observation are encoded into the shared latent space:

$$z_t = \text{enc}_\theta(o_t), \qquad z_g = \text{enc}_\theta(o_g) \tag{17}$$

No planning happens in pixel space; everything that follows operates entirely in $\mathbb{R}^d$.

Level	Action source	Requires search?	Semantic scale
3 — Strategic	$\text{ID}^{(3)}(z_t, z_g)$	No — single forward pass	Task-level intent
2 — Tactical	$\text{ID}^{(2)}(z_t, \hat{z}^{(2)}_{t+k_1})$	No — single forward pass	Sub-goal / waypoint
1 — Reactive	CEM over $a_{1:k_1}$	Yes — short-horizon	Motor control

Table 1: Action source at each level during inference. CEM is confined to Level 1.

6.3 Phase 1: Level 3 — Strategic Extrapolation (Single Forward Pass)

The first planning step produces a long-range anchor $\hat{z}^{(3)}_{t+k_2}$ in a single deterministic forward pass:

1. Query the Strategic ID model using z_g as a proxy for the true distant-future state:

$$\tilde{a}^{(3)}_t = \text{ID}^{(3)}(z_t,\ z_g) \tag{18}$$

2. Produce the long-range anchor:

$$\hat{z}^{(3)}_{t+k_2} = \text{pred}^{(3)}\Big(z_t,\ \tilde{a}^{(3)}_t\Big) \tag{19}$$

No CEM is required here. The ID model directly answers: *"Given where I am and where I need to end up, what high-level maneuver bridges them?"* The predictor then materializes the corresponding latent anchor state.

6.4 Phase 2: Level 2 — Tactical Interpolation (Single Forward Pass)

With z_t and $\hat{z}^{(3)}_{t+k_2}$ both known, the Level 2 predictor interpolates a midpoint:

1. Query the Tactical ID model. During planning, the true midpoint z_{t+k_1} is not available; the Level 2 ID model is queried using the Level 3 anchor $\hat{z}^{(3)}_{t+k_2}$ as a temporary proxy, or alternatively initialized from the midpoint between z_t and z_g in latent space:

$$\tilde{a}^{(2)}_t = \text{ID}^{(2)}\Big(z_t,\ \hat{z}^{(3)}_{t+k_2}\Big) \tag{20}$$

2. Produce the tactical midpoint, heavily constrained by the Level 3 anchor:

$$\hat{z}^{(2)}_{t+k_1} = \text{pred}^{(2)}\Big(z_t,\ \hat{z}^{(3)}_{t+k_2},\ \tilde{a}^{(2)}_t\Big) \tag{21}$$

No CEM is required here either. The Level 3 anchor acts as a gravitational constraint: the midpoint cannot drift arbitrarily far because the model knows the trajectory must pass through $\hat{z}^{(3)}_{t+k_2}$ at the later time.

6.5 Phase 3: Level 1 — Short-Horizon CEM (Anchored Search)

This is the *only* point in the pipeline where search is performed. CEM optimizes the physical action sequence $a_{1:k_1}$ over the short horizon $[t, t{+}k_1]$, with the rollout anchored to $\hat{z}^{(2)}_{t+k_1}$ from above:

Algorithm 1 Short-Horizon CEM at Level 1

Input: z_t, midpoint anchor $\hat{z}^{(2)}_{t+k_1}$, iterations T, candidates N, elites K
Initialize $\mu_0 = \mathbf{0}$, $\Sigma_0 = I$ over $a_{1:k_1}$
for $i = 1, \ldots, T$ **do**
  Sample N candidate sequences: $\{a^{(n)}_{1:k_1}\}_{n=1}^{N} \sim \mathcal{N}(\mu_{i-1}, \Sigma_{i-1})$
  for each candidate n **do**
    Roll out in latent space using $\text{pred}^{(1)}$, guided by $\hat{z}^{(2)}_{t+k_1}$:

$$\hat{z}^{(1)}_{t+j} = \text{pred}^{(1)}\left(\hat{z}^{(1)}_{t+j-1},\ \hat{z}^{(2)}_{t+k_1},\ a^{(n)}_j\right), \quad j = 1, \ldots, k_1$$

    Evaluate cost: $C^{(n)} = \left\| \hat{z}^{(1)}_{t+k_1} - \hat{z}^{(2)}_{t+k_1} \right\|_2^2$
  end for
  Select top-K elites by lowest cost
  Update: $\mu_i \leftarrow \text{mean(elites)}$, $\Sigma_i \leftarrow \text{var(elites)}$
end for
return $a^*_{1:k_1} \leftarrow \mu_T$ (best action sequence)

Key difference from base LeWM. CEM now searches over a short horizon of length $k_1 \ll H$ rather than the full planning horizon H. The cost function targets the *midpoint anchor* $\hat{z}^{(2)}_{t+k_1}$ rather than the distant goal z_g directly. This yields two compounding benefits: (1) the search space is k_1/H times smaller, requiring far fewer iterations to converge; and (2) the cost landscape is dramatically smoother, because the midpoint anchor is geometrically close to achievable trajectories.

6.6 Phase 4: Execute with Model Predictive Control

After CEM converges, we adopt the standard receding-horizon MPC strategy:

1. Execute only the first $K_{\text{exec}} \leq k_1$ physical actions from the optimal sequence $a^*_{1:k_1}$.
2. Re-observe the real environment: obtain $o_{t+K_{\text{exec}}}$.
3. Re-encode: $z_{t+K_{\text{exec}}} = \text{enc}_\theta(o_{t+K_{\text{exec}}})$.
4. Re-run the full top-down planning pipeline (Phases 1–3) from the updated state.

The hierarchical structure amplifies the benefit of MPC: even if low-level execution drifted due to environment noise, the Level 3 anchor $\hat{z}^{(3)}_{t+k_2}$ is immediately recomputed from the corrected state, re-orienting the tactical and reactive layers simultaneously.

6.7 Full Inference Flow (Summary)

7 Comparison to Base LeWM and Bottom-Up Hierarchies

8 Summary of Key Advantages

Zero compounding error at high levels. By predicting the distant future first and interpolating backward, the hierarchy ensures that every prediction at Levels 2 and 3 is constrained by a known target. Error accumulation is fully eliminated for these levels; Level 1 accumulates errors only over the short window $[t, t+k_1]$.

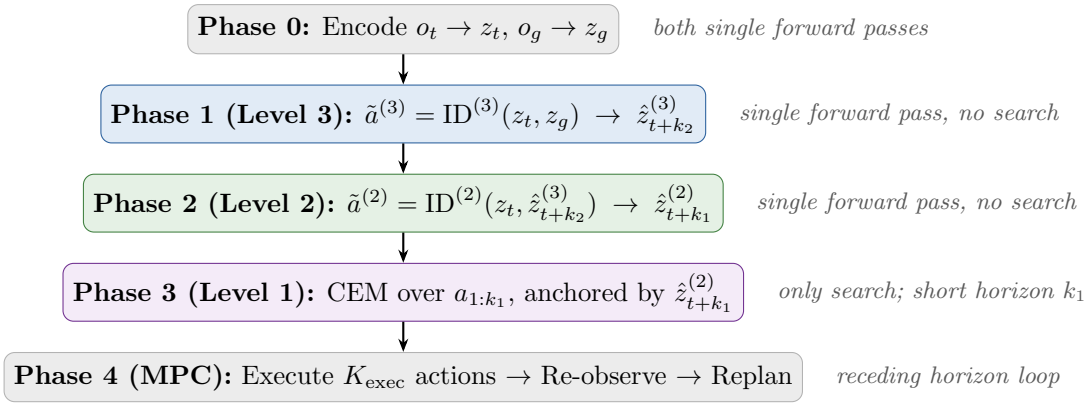

Figure 2: Full inference pipeline. Only Phase 3 requires iterative optimization; Phases 1 and 2 are single deterministic forward passes through the ID and predictor networks.

Property	**Base LeWM**	**Bottom-Up Hierarchy**	**Ours (Top-Down)**
Error accumulation	Compounds with H	Compounds at each level	Anchored at k_2, interpolated
Long-horizon reliability	Degrades with H	Degrades across levels	Stable; Level 1 always anchored
Latent spaces	1 (flat)	≥ 3 (separate)	1 (flat, shared)
SIGReg applications	1	≥ 3	1
Action abstraction	One level	Multiple (implicit)	3 levels via ID models
CEM horizon	Full H	Full H (per level)	Short k_1 only
CEM search space	$\mathbb{R}^{H \times d_a}$	$\mathbb{R}^{H \times d_a}$	$\mathbb{R}^{k_1 \times d_a}$
Planning speed	Fast	Slower (multi-tower)	Fastest (ID replaces high-level CEM)
Training complexity	2 terms, 1 param	≥ 6 terms, many params	3 terms, 2 params

Table 2: Comparison of planning architectures. H is the full planning horizon; $k_1 \ll H$ is the short CEM horizon.

Drastically smaller CEM search space. In base LeWM, CEM must search over the full H-step action sequence. In our proposal, it searches only over $k_1 \ll H$ steps. This reduces the dimensionality of the search problem by a factor of H/k_1, allowing CEM to converge faster and more reliably.

Smoother cost landscape for CEM. The Level 1 CEM is anchored to a nearby midpoint $\hat{z}^{(2)}_{t+k_1}$ rather than the distant goal z_g. The cost surface $\|\hat{z}^{(1)}_{t+k_1} - \hat{z}^{(2)}_{t+k_1}\|_2^2$ is far smoother and less multimodal than the base LeWM cost $\|\hat{z}^{(1)}_H - z_g\|_2^2$, leading to better optimization.

Incredibly simple state space. A single shared latent space eliminates the risk of independent representation collapse at multiple levels. A single application of SIGReg ensures the entire hierarchy's stability.

Minimal training complexity. The total objective adds only one new hyperparameter (α) over base LeWM, while gaining three levels of temporal abstraction. Compared to competing methods with six or more hyperparameters and complex multi-term losses, this is a substantial simplification.

Amplified MPC benefit. The hierarchical recomputation of the Level 3 anchor at each MPC replanning step immediately re-orients the entire hierarchy, making the agent more robust to execution noise than a flat MPC agent.

9 Semantic Interpretation: A Worked Example

To ground the abstract hierarchy in intuition, we trace inference through the robotic manipulation task: a robot arm must move a cup from the center of a table to the right side. All levels share the same present state z_t, encoded by the single shared encoder. What differentiates them is their *temporal scope*, not their representational format.

Initial Configuration

At time t, the system is in the following state:

- The robot arm is positioned on the left side of the table.
- The cup is located at the center of the table.

All levels of the hierarchy operate on this same initial state z_t.

Level 3 — Strategic (Long Horizon: $t \to t + k_2$)

Question answered: *What is the final outcome?*

Time	World state
t	Cup is at the center of the table
$t + k_2$	Cup is on the right side of the table

Semantic content of the Level 3 anchor $\hat{z}^{(3)}_{t+k_2}$: The cup has been fully relocated. The arm may be anywhere post-grasp. This level has no concept of joint angles, grasp configurations, or intermediate arm positions. It encodes only the coarse relational fact: object displacement at the task scale.

Level 3 does not ask *how* the cup moved; it only establishes *that* it did.

Level 2 — Tactical (Mid Horizon: $t \to t + k_1$)

Question answered: *What intermediate state is required to make the final outcome achievable?*

Time	World state
t	Arm is on the left; cup is at the center
$t + k_1$	Arm is aligned above the cup, ready to grasp

Semantic content of the midpoint $\hat{z}^{(2)}_{t+k_1}$: The arm has been repositioned into a pre-grasp configuration. The cup has not yet moved. This level encodes sub-goals and staging waypoints: it understands that grasping requires approach, and approach requires repositioning—but it knows nothing about the individual joint torques that will execute the approach.

Crucially, Level 2 could not have predicted this midpoint without the Level 3 anchor: knowing the cup must end up on the right side is what makes "hovering above the cup" the correct midpoint, rather than, say, moving the arm to an arbitrary position.

Level 1 — Reactive (Immediate Horizon: $t \to t+1$)

Question answered: *What exact physical action should be taken right now?*

Time	World state
t	Arm is slightly away from the cup
$t+1$	Arm has moved one step closer to the cup

Semantic content of the next-step prediction $\hat{z}^{(1)}_{t+1}$: A specific, small motor displacement—joint angles, end-effector velocity, gripper aperture. This level has no concept of the final goal. It only knows: (1) where it is now (z_t), and (2) where it needs to be within the next k_1 steps ($\hat{z}^{(2)}_{t+k_1}$). CEM optimizes the sequence of these fine-grained actions to reach the nearby midpoint.

The Decomposition Principle

Level	Core question	Example answer
3 — Strategic	What is the final outcome?	"Cup on the right"
2 — Tactical	What intermediate state is needed?	"Arm aligned above cup"
1 — Reactive	What action do I take right now?	"Extend joint 3 by 2°"

Without this decomposition, Level 1 alone would need to bridge "arm on the left" to "cup on the right in 200 steps"—an enormous credit assignment problem. The hierarchy decomposes this into three tractable sub-problems, each with a clear semantic scope, each constrained from above.

10 Ablation Studies

The following ablations isolate the contribution of each architectural component. Each ablation removes or replaces exactly one design choice while holding everything else constant, enabling clean causal attribution.

10.1 Ablation 1: Remove Level 3 (No Strategic Anchor)

Setup. Train and plan with Levels 1 and 2 only. Level 2 predicts the midpoint from z_t alone, without any long-range anchor:

$$\hat{z}^{(2)}_{t+k_1} = \text{pred}^{(2)}\Big(z_t,\ \tilde{a}^{(2)}_t\Big)$$

What this tests. Whether the Level 3 anchor is necessary, or whether Level 2 can discover good midpoints on its own.

Expected result. Significant degradation on long-horizon tasks ($H > k_2$). Without the distant anchor, Level 2 will produce midpoints that are locally plausible but globally inconsistent with the goal, leading Level 1 CEM to converge to the wrong region.

Would invalidate the proposal if: Performance is equivalent to the full model. This would suggest Level 3 adds no planning signal beyond what Level 2 can infer independently.

10.2 Ablation 2: Remove Level 2 (No Tactical Midpoint)

Setup. Train and plan with Levels 1 and 3 only. Level 1 CEM operates anchored directly to the Level 3 distant anchor:

$$C^{(n)} = \|\hat{z}^{(1)}_{t+k_1} - \hat{z}^{(3)}_{t+k_2}\|_2^2$$

What this tests. Whether the intermediate midpoint is necessary, or whether CEM can bridge from the current state to the distant anchor directly.

Expected result. CEM convergence degrades, especially when k_2 is large. A midpoint at distance k_1 is much easier to reach in short-horizon CEM than an anchor at distance k_2. Removing Level 2 effectively reintroduces the long-horizon CEM problem the hierarchy was designed to avoid.

Would invalidate the proposal if: CEM performs equally well targeting the distant anchor directly. This would suggest Level 2 is redundant overhead.

10.3 Ablation 3: Replace ID Models with CEM at All Levels (Flat CEM Baseline)

Setup. Remove all ID models. Run CEM jointly over $(a_{1:H}, \tilde{a}^{(2)}, \tilde{a}^{(3)})$ simultaneously—the full action space across all three levels—as in base LeWM but with the hierarchical predictors.

What this tests. Whether the ID models provide a meaningful shortcut, or whether CEM can find the same solutions by searching over all levels simultaneously.

Expected result. This baseline is strictly slower (larger search space) and likely less reliable (the joint search space is higher-dimensional and more multimodal). The hierarchical CEM cost landscape is harder to navigate than the anchored Level 1 cost.

Would invalidate the proposal if: Joint CEM outperforms the hybrid ID+CEM scheme. This would suggest the ID models introduce harmful approximation errors that hurt planning.

10.4 Ablation 4: Apply SIGReg at Every Level Separately

Setup. Add separate SIGReg terms for the outputs of $\text{pred}^{(2)}$ and $\text{pred}^{(3)}$, treating each level's predicted states as a separate distribution to regularize.

What this tests. Whether the single shared SIGReg is sufficient, or whether the higher-level predictors need their own anti-collapse regularization.

Expected result. No significant improvement, and potentially slight degradation (over-regularization, training instability). Because all predictions live in the same flat latent space already constrained by the base SIGReg, adding redundant regularization should be neutral to harmful.

Would invalidate the proposal if: Multi-level SIGReg significantly improves performance. This would imply the higher-level predictors can develop their own collapse modes despite operating in a regulated base space.

10.5 Ablation 5: Varying k_1 and k_2 (Temporal Scale Sensitivity)

Setup. Fix the task horizon H and vary (k_1, k_2) across configurations, e.g., $(k_1, k_2) \in \{(5, 20), (10, 40), (20, 80)\}$.

What this tests. How sensitive performance is to the choice of temporal scales, and whether a "sweet spot" ratio k_2/k_1 exists.

Expected result. A Goldilocks region: if k_1 is too small, Level 1 CEM has no room to maneuver; if k_2 is too large, the Level 3 predictor must extrapolate too far, producing unreliable anchors. A ratio of $k_2/k_1 \approx 3$–5 is a reasonable prior based on the literature on temporal abstraction.

11 Probing Experiments: Success and Failure Criteria

This section defines what a successful implementation looks like in practice—and what warning signs indicate a deeper problem. These criteria should be evaluated before running full task benchmarks.

11.1 Probing Experiment 1: Latent Space Geometry

Goal. Verify that the shared latent space is well-structured and that temporal distance corresponds to geometric distance.

Protocol.

1. Encode a trajectory $\{z_t, z_{t+1}, \ldots, z_{t+k_2}\}$ from a held-out episode.
2. Compute pairwise L_2 distances in latent space.
3. Plot distance as a function of temporal separation Δt.

What to expect (success). Distance should be a monotonically increasing function of Δt, with roughly linear growth in the early steps and possible saturation at large Δt. The anchor $\hat{z}^{(3)}_{t+k_2}$ should be geometrically closer to the true z_{t+k_2} than to z_t.

Red flag. If the distance function is flat or non-monotonic, the encoder has not learned a temporally coherent space—the hierarchy will produce meaningless anchors.

11.2 Probing Experiment 2: ID Model Fidelity

Goal. Verify that the ID models correctly infer macro-actions from state pairs, and that these actions produce accurate predictions when fed to the corresponding predictors.

Protocol.

1. Sample ground-truth state pairs (z_t, z_{t+k_2}) from held-out data.
2. Compute $\tilde{a}^{(3)} = \mathrm{ID}^{(3)}(z_t, z_{t+k_2})$.
3. Feed to $\mathrm{pred}^{(3)}$: compute $\hat{z}^{(3)}_{t+k_2}$.
4. Measure reconstruction error $\|\hat{z}^{(3)}_{t+k_2} - z_{t+k_2}\|_2^2$.

Repeat for Level 2.

What to expect (success). Reconstruction error at Level 3 should be substantially lower than the distance between random state pairs. A useful threshold: $\|\hat{z}^{(3)}_{t+k_2} - z_{t+k_2}\|_2 < 0.5 \cdot \|z_{t+k_2} - z_t\|_2$.

Red flag. If reconstruction error is similar to baseline (random anchor), the ID model has failed. The most common failure mode is the ID model ignoring the action representation entirely and the predictor learning to output a fixed average state—a soft collapse.

11.3 Probing Experiment 3: Anchor Quality During Planning

Goal. Verify that the Level 3 anchor produced at *inference time* (using z_g as a proxy) is semantically meaningful, not just a memorized training artifact.

Protocol.

1. For N held-out goal images o_g, compute $\tilde{a}^{(3)} = \mathrm{ID}^{(3)}(z_t, z_g)$ and $\hat{z}^{(3)}_{t+k_2}$.
2. Decode $\hat{z}^{(3)}_{t+k_2}$ to pixel space using a lightweight decoder (trained separately, not part of the planning pipeline).

3. Visually or quantitatively compare decoded anchors to the true goal images.

What to expect (success). Decoded anchors should visually resemble the goal configuration—correct object position, arm pose in the vicinity of the goal. Perfect reconstruction is not expected; the anchor should capture the correct *region* of state space.

Red flag. Decoded anchors that resemble the current state z_t more than z_g indicate the ID model is ignoring the goal signal. Anchors that are blurry averages of all training states indicate collapse.

11.4 Probing Experiment 4: CEM Convergence Speed at Level 1

Goal. Verify that Level 1 CEM converges faster and to lower cost than flat base LeWM CEM, due to the shorter horizon and smoother cost landscape.

Protocol.

1. For matched tasks, run Level 1 CEM (anchored, horizon k_1) and base LeWM CEM (flat, horizon H) with the same $N = 300$, $K = 30$ budget.

2. Record cost C as a function of CEM iteration i.

3. Compare convergence curves and final costs.

What to expect (success). Level 1 CEM should converge to a lower cost in fewer iterations. The variance across random seeds should be lower (smoother landscape). A concrete target: 50% reduction in iterations needed to reach within 10% of the final cost.

Red flag. If Level 1 CEM converges to a *higher* cost than flat CEM, the midpoint anchor may be placing the CEM search in the wrong region of action space—the anchor is pulling CEM away from the true optimal plan.

11.5 Probing Experiment 5: Hierarchy Consistency Check

Goal. Verify that the three levels produce a geometrically coherent trajectory in latent space, i.e., the midpoint genuinely lies between the current state and the anchor.

Protocol.

1. Run the full inference pipeline to obtain z_t, $\hat{z}^{(2)}_{t+k_1}$, $\hat{z}^{(3)}_{t+k_2}$, and z_g.

2. Compute the "interpolation ratio":

$$\rho = \frac{\|\hat{z}^{(2)}_{t+k_1} - z_t\|_2}{\|\hat{z}^{(3)}_{t+k_2} - z_t\|_2}$$

3. Compare ρ to the expected geometric ratio k_1/k_2.

What to expect (success). $\rho \approx k_1/k_2$, indicating the midpoint is geometrically proportional along the trajectory from current state to anchor. The midpoint should lie roughly "between" z_t and $\hat{z}^{(3)}_{t+k_2}$ in the direction of z_g.

Red flag. $\rho \gg k_1/k_2$ indicates the midpoint has overshot the anchor. $\rho \approx 0$ indicates the midpoint has collapsed to the current state. Either failure mode means Level 2 is not learning to interpolate.

11.6 Summary: At-a-Glance Criteria

Experiment	**Success looks like**	**Failure looks like**
Latent geometry	Monotonic distance vs. Δt	Flat or chaotic distance curve
ID fidelity	Reconstruction error $< 0.5\times$ baseline	Error similar to random anchors
Anchor quality (planning)	Decoded anchors near goal region	Anchors resemble z_t, not z_g
CEM convergence	Faster, lower-variance convergence	Level 1 CEM worse than flat CEM
Hierarchy consistency	$\rho \approx k_1/k_2$	$\rho \approx 0$ or $\rho \gg k_1/k_2$

Table 3: Summary of probing experiments and their success/failure criteria. These should be evaluated before any full task benchmark to diagnose whether the hierarchy has been learned correctly.